

Python-Based Engine for Music Scripting Usability Survey

* Required

Python-Based Engine for Music Scripting Usability Survey

Information About Participation

The aim of this research is to gain valuable user insight into the performance and usability of a prototyped engine for music making through coding. The study aims to evaluate how effectively users can understand, and interact with the system, as well as to identify any usability challenges or areas for improvement. Findings from this study will be used to inform further development of the engine and to assess its suitability for educational use in supporting both programming and musical learning. The research is conducted by university undergraduate student Fraser McCarter with supervision from Dr Debbie Meharg. The results of this research will be used for an MA honours project report which will be made accessible through the project's online publicly accessible GitHub repository.

We appreciate your interest in participating in this set of online tasks and accompanying questionnaire. We would like to invite you to take part if you are interested in music, coding, or creative digital tools, regardless of prior experience level. Please take some time to read the following information carefully before agreeing to participate (if you wish to) by ticking the consent boxes at the bottom of this page. Your participation in the survey is entirely voluntary, and you can opt out at any stage by closing and exiting the browser.

If you decide to take part, you will be asked to complete a short set of guided tasks using the prototype engine or watch a short video demonstration of these tasks. These tasks are designed to evaluate how easily users can understand and interact with the system. After completing each task, you will be asked to reflect on your experience by answering a short questionnaire about task difficulty, clarity, and overall usability. Your answers will help us to gain valuable insight on user interaction with the system along with informing potential future design and development. The questionnaire and associated online tasks should take approximately **10-20 minutes to complete**. Please be aware that after completing the questionnaire, you will not be able to withdraw your data.

Your answers will be treated confidentially and the information you provide will remain anonymous in any scholarly outputs. All data generated by the study will be securely stored at the University of Aberdeen and retained in accordance with the University of Aberdeen Research Data Management Policy for 5 years after the project end date. Your data can only be accessed by the primary investigator. You can also consult the University's Privacy Notice for Research Participants for further information on how we manage your data.

The research has been approved by the School of Natural and Computing Sciences on the Day Month Year [To be added after ethical approval has been given]. If you have any concerns about the way in which the study has been conducted, you can contact the Course/Programme Coordinator, Dr Miles Everett, at miles.everett@abdn.ac.uk.

For further information, or if you have any queries, please contact the lead researcher Fraser McCarter at u02fm22@abdn.ac.uk.

1. Please read the statements below carefully and tick the box to confirm you have read and understood each statement and agree to participate in the project. By clicking Next you will then be given access to the questionnaire.

- I confirm that I am 18 years of age or over.
- I agree to have my data securely collected, stored, and processed as part of this research under the conditions outlined in the participant information above.
- I agree to my anonymous data being used for publications and other scholarly means of disseminating the findings from the research project.

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☐ I confirm that I have read and understood the above statements and I consent to take part in the research study Prototyping a Python-Based Engine for Music Scripting and Algorithmic Composition under the outlined conditions (tick the box).

Python-Based Engine for Music Scripting - Preface

This survey consists of **five** sections:

- Programming and Musical Background
- Task One
- Task Two
- Task Three
- Overall System Review

Each section is expected to take **less than five minutes** to complete, with tasks taking a **maximum of ten minutes** to complete and evaluate . If tasks take longer than this to complete it is advised to halt the task at hand and write any issues / frustrations in the designated section at the bottom of the task page before moving on to the next task.

To complete the interactive survey you will need access to the following things:

- A laptop or computer with internet access
- A coding environment of your choice (VScode is recommended)
- A laptop or computer with speakers or an audio jack
- Headphones (optional)

To complete the video demo survey you will need access to the following things:

- A device to complete the survey on (e.g. mobile, laptop, computer)
- A device with speakers or an audio jack
- Headphones (optional)

Python-Based Engine for Music Scripting Usability Survey - Background Questions

2. How much experience do you have with programming? (1 - none, 5 - proficient in multiple languages) *

1	2	3	4	5
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3. Which programming languages have you used if any? (select all that apply)

- ☐ Python
- ☐ Java
- ☐ C / C++
- ☐ Javascript
- ☐ Block-based environments (e.g. Scratch)
- ☐ Other

4. Have you previously written code in a text-based programming language? *

- ☐ Yes
- ☐ No

5. How confident do you feel about writing code without visual or block-based support? (1 - not at all, 5 - very confident) *

1	2	3	4	5
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6. Which of the following best describe your musical experience? *

- ☐ None
- ☐ Hobbyist (e.g. casual playing / listening)
- ☐ Formal training (e.g. lessons, school, university)
- ☐ Professional or semiprofessional

7. How familiar are you with basic music theory concepts? (1 - not at all, 5 - very familiar) *

1	2	3	4	5
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8. Have you previously used any digital music tools? (select all that apply) *

- ☐ Digital audio workstations (e.g. Ableton, Logic, Pro Tools)
- ☐ Music notation software (e.g. Sibelius, Musescore)
- ☐ Music coding environments (e.g. Sonic Pi, Pure Data)
- ☐ None of the above

9. Have you previously combined music and coding in any way? *

- ☐ Yes
- ☐ No

10. What motivated you to take part in this study? *

- ☐ Interest in programming
- ☐ Interest in music
- ☐ Interest in music and coding
- ☐ Curiosity about the system
- ☐ Other

11. When learning something new, which approach do you generally prefer? *

- ☐ Visual interfaces
- ☐ Text-based systems with examples
- ☐ A mix of visual and text-based tools
- ☐ Not sure

Python-Based Engine for Music Scripting - Survey Type

If you have some experience using a coding environment then feedback on your use of the tool would be preferred. Please select the interactive survey.

If you do not have experience running code within a coding environment or do not have the means to access one right now then please select the video demo survey.

12. Survey type you wish to conduct *

☐ Interactive Survey

☐ Video Demo Survey

Python-Based Engine for Music Scripting Usability Survey - Task One(Interactive)

In a separate tab please open the link below, this will take you to the GitHub repository for the code. Download the code, extract it and open the folder in your chosen coding environment. Each task will be labelled with its own file in the side tab ([task1.py](#), [task2.py](#), etc.). Please select [task1.py](#) for this task.
<https://github.com/frasermccarter/CS4525-Joint-Honours-Project>

Task One

- Find the Eb note and change it to an E, then run the code to play the result.
- Now change the octave of the G note from an G3 to a G4, then run the code to play the result.
- Change the duration of the C note from 1 to 0.5, then run the code to play the result.
- Create a new A4 note (with a duration and velocity of your choosing) that plays after the G note, then run the code to play the result.

13. How easy was it to locate the Eb note in the code? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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14. How easy was it to modify the Eb note to an E? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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15. How easy was it to modify the octave of the G note? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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16. How easy was it to create an A note after the G note? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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17. How easy was it to alter note properties such as pitch, duration and velocity? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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18. How easy was it to play the sequence and receive an output? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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19. How confident did you feel in completing this task? (1 - not confident at all, 5 - very confident) *

1	2	3	4	5
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20. To what extent did this task help you understand how notes are represented in the system? (1 - not at all, 5 - very much) *

1	2	3	4	5
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21. How clear was the relationship between the code you edited and the sound produced? (1 - not at all, 5 - very much) *

1	2	3	4	5
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22. How easy was it to predict what would happen when you altered the code? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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23. How satisfying did you find completing this task? (1 - not satisfying at all, 5 - very satisfying) *

1	2	3	4	5
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24. Did you experience any frustration in completing this task? *

☐ Yes

☐ No

25. If yes, describe this frustration.

26. Are there additional comments you would like to make on this task?

Python-Based Engine for Music Scripting Usability Survey - Task One(Video Demo)

Please watch the video demo for task one below. Try to imagine as though you yourself are using the program and think about what it would be like to complete this task and interact with the system. The task at hand is shown in the video as well as listed below:

https://youtu.be/8FCgav_1HQU

Task One

- Find the Eb note and change it to an E, then run the code to play the result.
- Now change the octave of the G note from an G3 to a G4, then run the code to play the result.
- Change the duration of the C note from 1 to 0.5, then run the code to play the result.
- Create a new A4 note (with a duration and velocity of your choosing) that plays after the G note, then run the code to play the result.

27. How easy was it to locate the Eb note in the code? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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28. How easy was it to modify the Eb note to an E? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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29. How easy was it to modify the octave of the G note? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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30. How easy was it to create an A note after the G note? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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31. How easy was it to alter note properties such as pitch, duration and velocity? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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32. How easy was it to play the sequence and receive an output? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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33. How confident would you feel in completing this task if the environment was setup for you? (1 - not confident at all, 5 - very confident) *

1	2	3	4	5
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34. To what extent did this task help you understand how notes are represented in the system? (1 - not at all, 5 - very much) *

1	2	3	4	5
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35. How clear was the relationship between the code you saw being edited and the sound produced?
(1 - not at all, 5 - very much) *

1	2	3	4	5
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36. How easy was it to predict what would happen when the code was altered? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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37. How satisfying would you find completing this task if you were to do it on your own? (1 - not satisfying at all, 5 - very satisfying) *

1	2	3	4	5
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38. Are there additional comments you would like to make on this task?

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Python-Based Engine for Music Scripting Usability Survey - Task Two(Interactive)

In a separate tab please open the link below, this will take you to the GitHub repository for the code. Download the code, extract it and open the folder in your chosen coding environment. Each task will be labelled with its own file in the side tab ([task1.py](#), [task2.py](#), etc.). If you have already done this step then ignore and continue. Please select [task2.py](#) for this task. <https://github.com/frasermccarter/CS4525-Joint-Honours-Project>

Task Two

- Locate the arpeggio function and change the chord from a C major (Cmaj) to a C minor (Cmin), then run the code to play the result.
- Change the arpeggio from playing an ascending sequence (up) to a descending sequence (down), then run the code to play the result.
- Fill in the parameters for the random walk function, then run the code to play the result.

You must fill in:

- start_pitch: The starting pitch of the random walk (e.g. 'C4', 60 (midi), 440.0 (hz) etc.).
- num_notes: The total number of notes to generate in the random walk (e.g. 4, 5, 16 etc.).
- max_step: The maximum step size (in semitones) for the random walk. A max_step of 2 means that each note can be at most 2 semitones apart from each other.
- duration: The duration of each note in beats (e.g. 0.5 for quaver, 1 for crochet, etc.).
- scale_type: The scale to constrain the random walk to (e.g. "major", "minor", "chromatic"). If left unspecified (scale_type=None), the random walk will be generated without any scale constraints, meaning it can include any pitch.

39. How easy was it to locate the arpeggio function in the code? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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40. How easy was it to change the arpeggiated chord from C major to C minor? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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41. How easy was it to change the arpeggio from ascending to descending? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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42. How well did you understand the arpeggio function before you modified it? (1 - not at all, 5 - very well) *

1	2	3	4	5
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43. How well did you understand the arpeggio function after you modified it? (1 - not at all, 5 - very well) *

1	2	3	4	5
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44. How easy was it to modify the code for, and use, the random walk function? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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45. How confident did you feel making changes to the functions without breaking them? (1 - not confident, 5 - very confident) *

1	2	3	4	5
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46. How clear was the relationship between the code changes you made and the musical output you heard? (1 - not clear, 5 - very clear) *

1	2	3	4	5
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47. Did you experience any frustration in completing this task? *

☐ Yes

☐ No

48. If yes, describe this frustration.

49. Are there additional comments you would like to make on this task?

Python-Based Engine for Music Scripting Usability Survey - Task Two (Video Demo)

Please watch the video demo for task two below. Try to imagine as though you yourself are using the program and think about what it would be like to complete this task and interact with the system. The task at hand is shown in the video as well as listed below:

<https://youtu.be/2kGDQyuzPtc>

Task Two

- Locate the arpeggio function and change the chord from a C major (Cmaj) to a C minor (Cmin), then run the code to play the result.
- Change the arpeggio from playing an ascending sequence (up) to a descending sequence (down), then run the code to play the result.
- Fill in the parameters for the random walk function, then run the code to play the result.

You must fill in:

- start_pitch: The starting pitch of the random walk (e.g. 'C4', 60 (midi), 440.0 (hz) etc.).
- num_notes: The total number of notes to generate in the random walk (e.g. 4, 5, 16 etc.).
- max_step: The maximum step size (in semitones) for the random walk. A max_step of 2 means that each note can be at most 2 semitones apart from each other.
- duration: The duration of each note in beats (e.g. 0.5 for quaver, 1 for crochet, etc.).
- scale_type: The scale to constrain the random walk to (e.g. "major", "minor", "chromatic"). If left unspecified (scale_type=None), the random walk will be generated without any scale constraints, meaning it can include any pitch.

50. How easy was it to locate the arpeggio function in the code? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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51. How easy was it to change the arpeggiated chord from C major to C minor? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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52. How easy was it to change the arpeggio from ascending to descending? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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53. How well did you understand the arpeggio function before it was modified? (1 - not at all, 5 - very well) *

1	2	3	4	5
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54. How well did you understand the arpeggio function after it was modified? (1 - not at all, 5 - very well) *

1	2	3	4	5
---	---	---	---	---

55. How easy was it to modify the code for, and use, the random walk function? (1 - very hard, 5 - very easy) *

1	2	3	4	5
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56. How confident would you feel making changes to these functions without breaking them? (1 - not confident, 5 - very confident) *

1	2	3	4	5
---	---	---	---	---

57. How clear was the relationship between the code changes made and the musical output you heard? (1 - not clear, 5 - very clear) *

1	2	3	4	5
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58. Are there additional comments you would like to make on this task?

Python-Based Engine for Music Scripting Usability Survey - Task Three(Interactive)

Participants **are not** required to create an account or provide personal data on third-party platforms. Though not expected, this task may be skipped if the participant feels frustrated at any point.

In a separate tab please open the link below, this will take you to the GitHub repository for the code. Download the code, extract it and open the folder in your chosen coding environment. Each task will be labelled with its own file in the side tab ([task1.py](#), [task2.py](#), etc.). Please select [task3.py](#) for this task.

<https://github.com/frasermccarter/CS4525-Joint-Honours-Project>

Task Three

- Run the code and listen to its musical output.
- Export the musical output of the Python script as a MIDI file using: `export_midi("task3_output", tempo=120)`
- Once exported, upload the MIDI file to the online MIDI editor / player linked below:

<https://signalmidi.app/edit>

Please listen and compare the results to what you expect based on the code input and whether the MIDI output is an accurate representation of the code.

59. How easy was it to export the code output to a MIDI file? (1 - very hard, 5 - very easy)

1	2	3	4	5
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60. How easy was it to locate and upload the MIDI file into the online player? (1 - very hard, 5 - very easy)

1	2	3	4	5
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61. How clear were the instructions for completing this task? (1 - very unclear, 5 - clear)

1	2	3	4	5
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62. How much effort did this task require compared to the previous tasks? (1 - much more effort, 3 - about the same, 5 - much less effort)

1	2	3	4	5
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63. Did anything about the export or import process cause confusion or difficulty? (if not, enter 'no').

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64. Where there any differences between the live code output and the MIDI playback that you noticed? (if not, enter 'no').

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Python-Based Engine for Music Scripting Usability Survey - Task Three(Video Demo)

Please watch the video demo for task three below. Try to imagine as though you yourself are using the program and think about what it would be like to complete this task and interact with the system. The task at hand is shown in the video as well as listed below:

<https://youtu.be/Yf90T9oSnIl>

Task Three

- Run the code and listen to its musical output.
- Export the musical output of the Python script as a MIDI file using: `export_midi("task3_output", tempo=120)`
- Once exported, upload the MIDI file to the online MIDI editor / player linked below:

<https://signalmidi.app/edit>

Please listen and compare the results to what you expect based on the code input and whether the MIDI output is an accurate representation of the code.

65. How easy was it to export the code output to a MIDI file? (1 - very hard, 5 - very easy)

1	2	3	4	5
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66. How easy was it to locate and upload the MIDI file into the online player? (1 - very hard, 5 - very easy)

1	2	3	4	5
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67. How clear were the instructions for completing this task? (1 - very unclear, 5 - clear)

1	2	3	4	5
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68. How much effort did this task require compared to the previous tasks? (1 - much more effort, 3 - about the same, 5 - much less effort)

1	2	3	4	5
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69. Did anything about the export or import process cause confusion or difficulty? (if not, enter 'no').

70. Where there any differences between the live code output and the MIDI playback that you noticed? (if not, enter 'no').

Python-Based Engine for Music Scripting Usability Survey - Overall System Competance

This is the final section of the survey and consists of questions about the overall usability of the system and your personal satisfactions or frustrations in using it.

71. I think that I would use this system frequently. (1 - never, 5 - very often) *

1	2	3	4	5
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72. I thought the system was easy to use. (1 - very hard, 5 - very easy) *

1	2	3	4	5
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73. I think I would need the support of a technical person in order to use this system. (1 - not at all, 5 - definitely) *

1	2	3	4	5
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74. I thought the various functions of this program were well integrated. (1 - not at all, 5 - very well) *

1	2	3	4	5
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75. I thought there was too much inconsistency in the system. (1 - not at all, 5 - very much so) *

1	2	3	4	5
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76. I imagine most people would learn to use this system very quickly. (1 - not at all, 5 - very much so) *

1	2	3	4	5
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77. I found the system very cumbersome to use. (1 - not at all, 5 - very cumbersome) *

1	2	3	4	5
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78. I felt confident in the system. (1 - not at all, 5 - very confident) *

1	2	3	4	5
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79. I needed to learn a lot about the system before I could get going. (1 - not at all, 5 - very much so) *

1	2	3	4	5
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Python-Based Engine for Music Scripting Usability Survey - Conclusion

Thank you very much for taking the time to participate in this survey, if you are happy with the answer you have given you may submit them by pressing the **Submit** button below.

For further information, or if you have any queries, please contact the lead researcher Fraser McCarter at u02fm22@abdn.ac.uk.

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